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Investigating Noise Robust Training with tunable Nonlinearities in ReLU-based models

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Introduction

- Neural networks often perform well on clean datasets, but their accuracy drops significantly under real-world conditions involving noise or blur.
- To train a regular neural network with about 128, 64 **Hidden Layers**, the model tunes about **110,000 parameters**
- To investigate this problem, instead of training a standard neural network opted to use **ReLU Non-linear Activation Training (NLAF)**
- This method allows the model to train on the non-linearities which consist of about **200 parameters**
- The experiment was conducted on the MNIST Dataset, which is a collection of 60,000 images that consists of handwritten numbers from 1-9, and evaluated the **test accuracies** of the simulations by applying noise distortions.

Methodology

Model 1: The experiment was divided into two phases, **Phase 1** involved gathering data for a standard training model to compare that to the data of **Phase 2** which was where the nonlinearities specifically were trained, freezing the other parameters ; the weights and biases in the process.

Phase 1

Phase 2

Step 1
Train on Clean MNIST Dataset

Step 2
Test on Uniform Noisy Dataset

Step 3
Train on Randomly Generated Noisy Dataset

Step 4
Test on Uniform Noisy Set

Step 1
Train on Clean MNIST Dataset

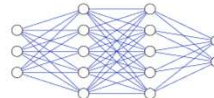
Step 2
Train on Uniform Noisy Dataset

Step 3
Train on Noisy Dataset with on ReLU nonlinearities Dense Layers Frozen

Step 4
Test on Uniform Noisy Dataset

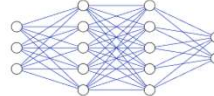
Model Performance

Clean Dataset



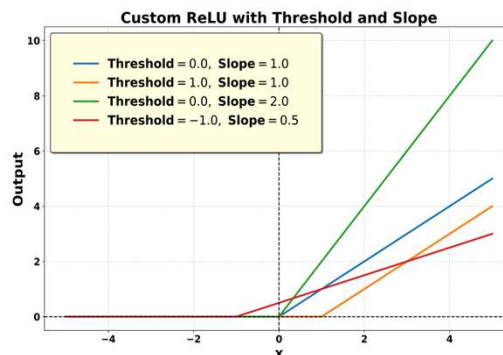
9

Noisy Dataset



7

Model 2: The two images are pulled out from the MNIST dataset, and the model classifies the very distorted image of the number 7 as a 9 when trained a **clean dataset**. Comparatively when trained on a **sample of noisy data**, the model can correctly classify the image.



Model 3: Graphical Representation of the **tunable nonlinearities** and what it does to each input of a neural network

Noise Type	Params	Test Accuracy	Test Accuracy (NAT)	Test Accuracy (NLAF)
Salt	P = 0.5	27%	97%	67%
Gaussian Noise	$\sigma = 150$	43%	97%	80%
Blur	(5,5)	90%	96%	95%
Gaussian Blur	(5,5)	95%	97%	97%

Table 1: This data involves the 4 noise types that we used in our experiment.. The collected test accuracies of the clean dataset, Noise-Aware Training (NAT), and **Non-linear activation function, (NLAF)** are shown above. By only just Training 0.2% of the weights the accuracy is almost close to NAT.

Conclusion & Discussion

Dense neural networks are vulnerable to input distortions such as noise and blur, as seen by the results. Noise-aware training improves accuracy on corrupted inputs, by training them on an range of noisy data so the model can adapt to the noisy testing images. Using **trainable activation functions** enables adaptation without touching model weights, which is much less power consuming. This strategy provides a new pathway for making models more resilient while minimizing energy and compute requirements. This can also be potentially implemented in CNN and in photonics.

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